HND (IT) Year 2, Semester 2

Project and Professional Aspects of IT

Project management Tools and SDLC models

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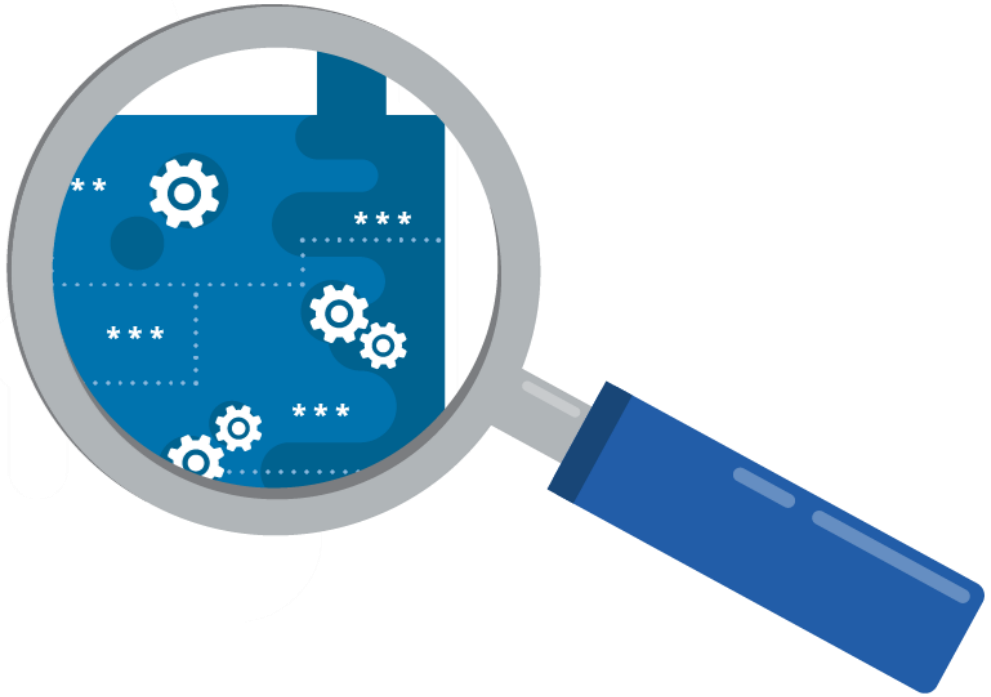
Overview of the lecture

- Project Management and related tools.
- Project Evaluation and review Technique Charts
- SLDC



Project Management Tools?

Project management tools are software or platforms designed to help teams plan, execute, monitor, and complete projects efficiently. These tools often include features for task management, resource allocation, communication, and reporting.

PM Tools – Online Tools

Wrike

Jira Software

asana

Bitrix24

ZOHO PROJECTS

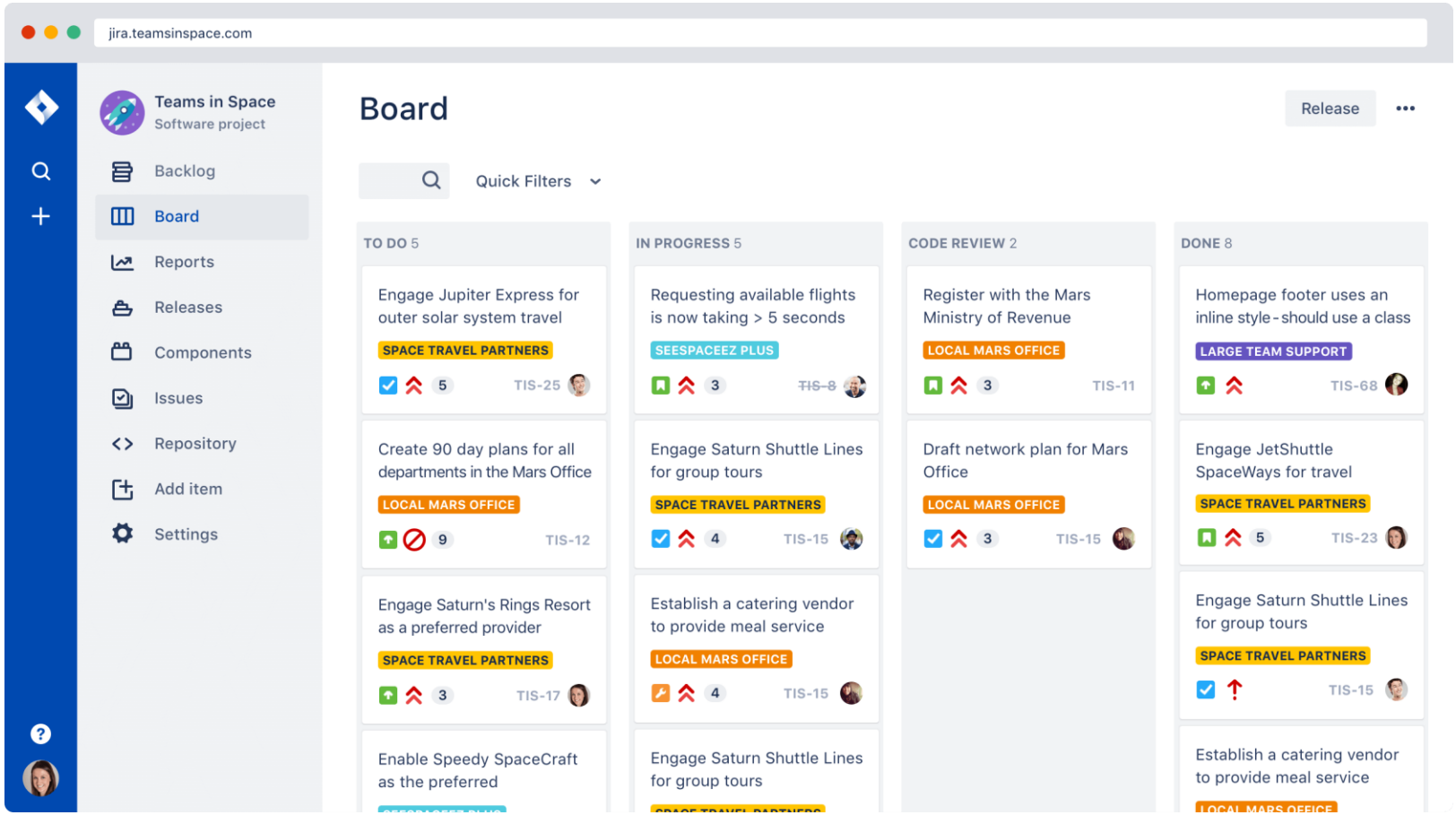
ClickUp

meister task

Trello

HYGGER
A cozy home for your product

Project Management tools - Example



Version Control System?

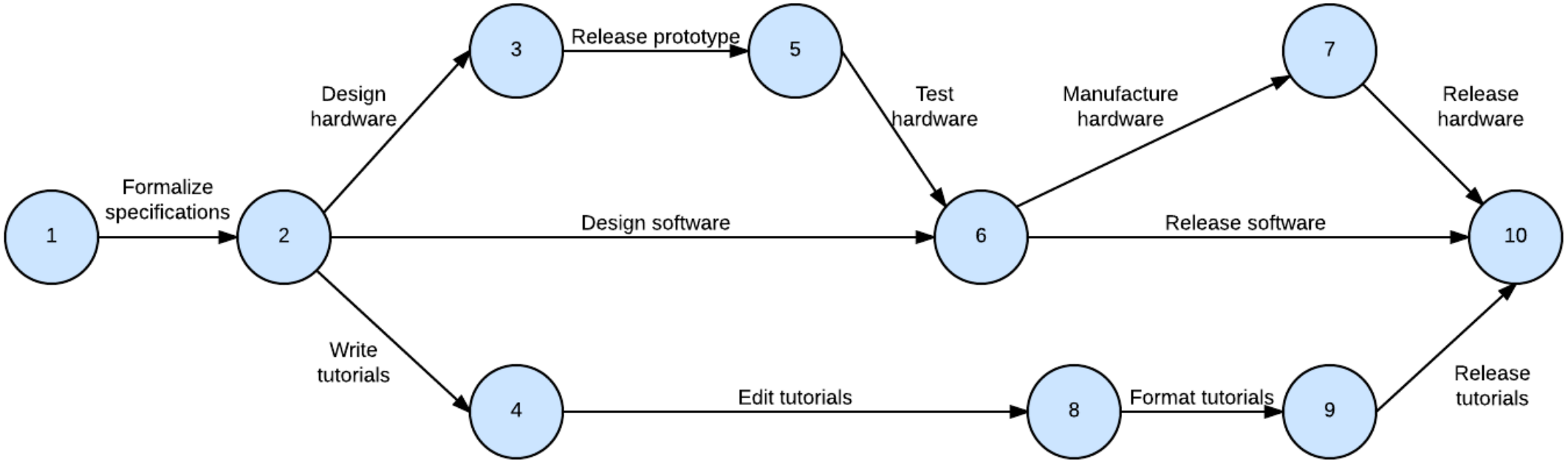
Version Control System (VCS) is a software tool that helps software developers manage changes to source code or other files over time. It tracks the history of changes, allowing multiple developers to collaborate on a project while maintaining control over different versions of files.

git

mercurial

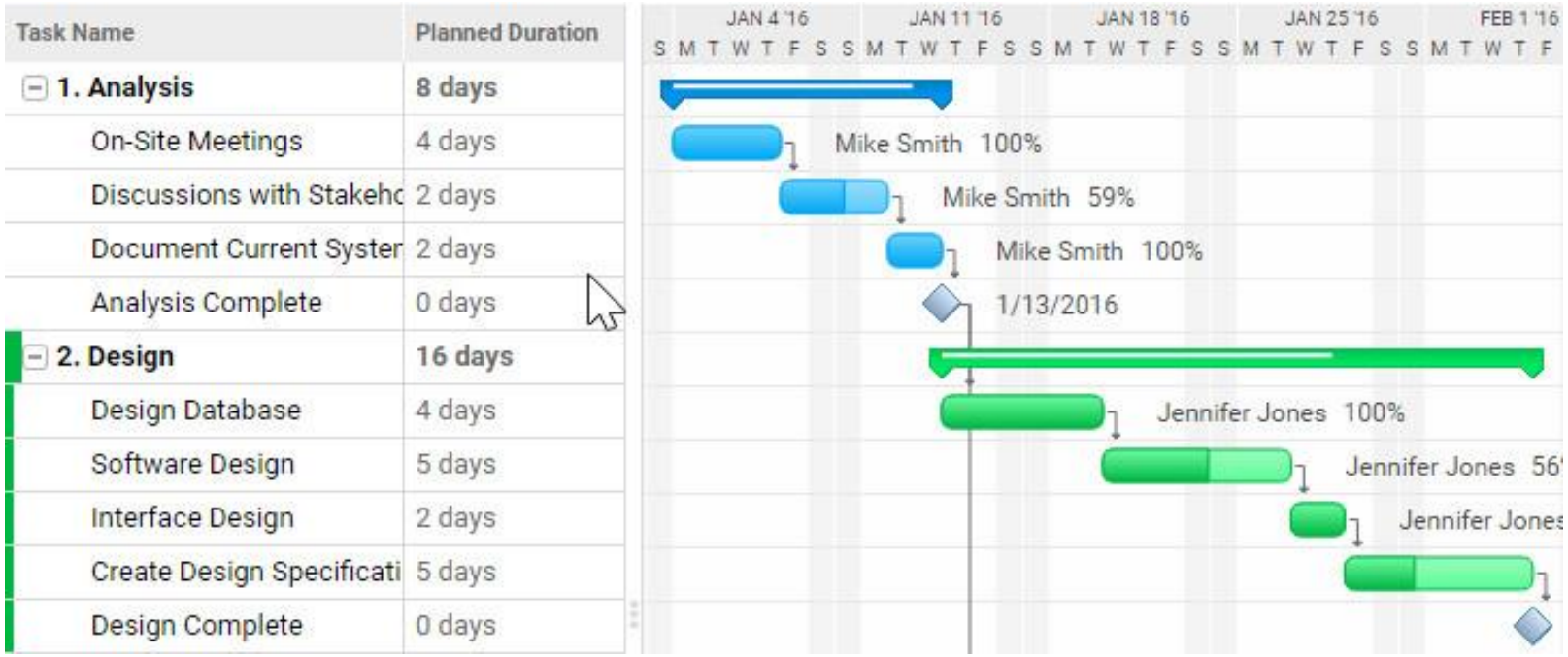
ATLASSIAN Bitbucket

Program Evaluation Review Technique Charts (PERT charts)



A project management tool used to schedule, organize and coordinate tasks within a project.

Gantt Charts



teamgantt

ClickUp

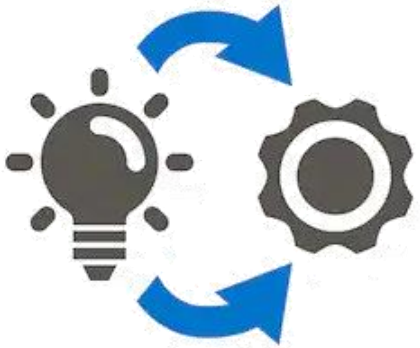
Software Development Life Cycle

The Software Development Lifecycle is a systematic approach to build software with quality and correctness

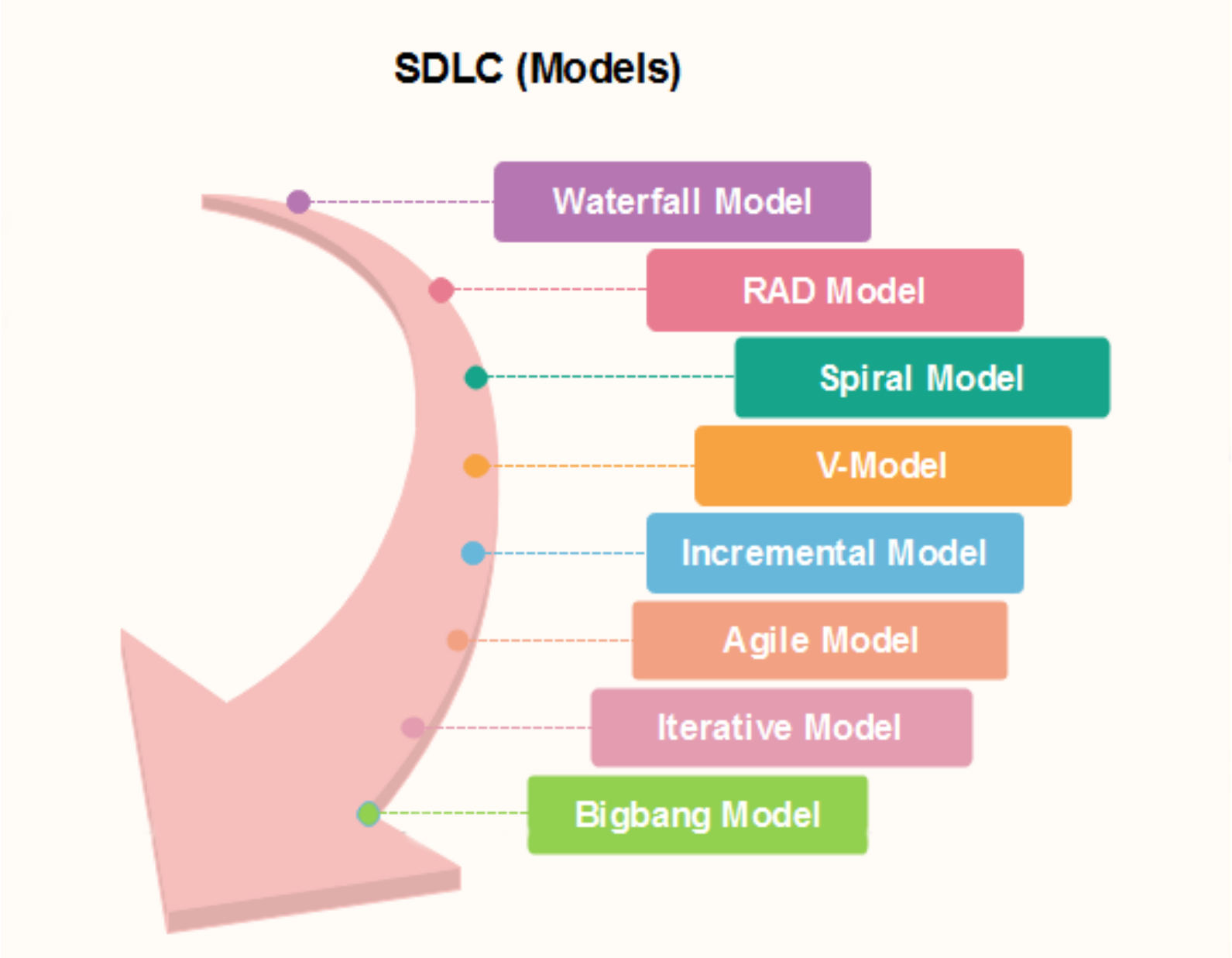
Why SDLC ?

SDLC Phases

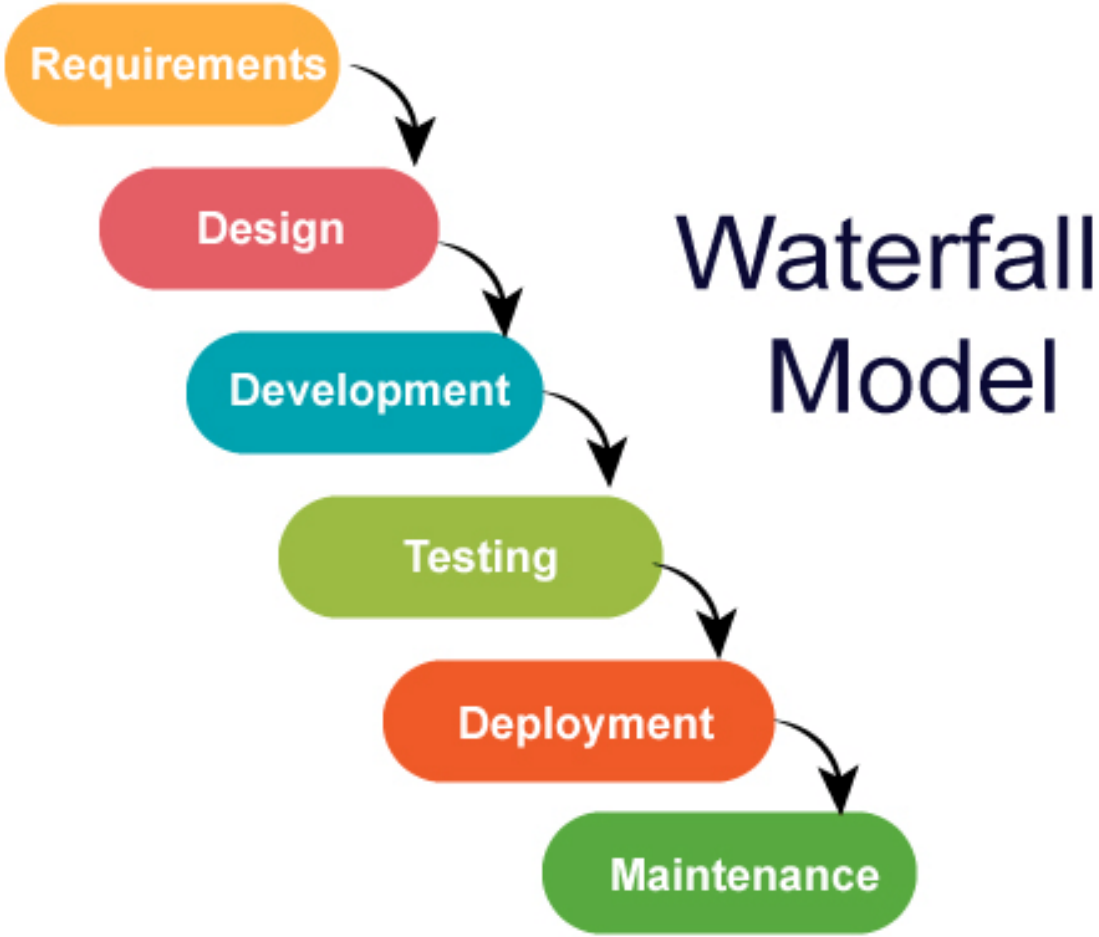
Phase 1: Requirement Gathering
Phase 2: Analysis and Feasibility study
Phase 3: Design
Phase 4: Implementation
Phase 5: Testing
Phase 6: Installation/Deployment
Phase 7: Maintenance



SDLC models


SDLC (Models)
Waterfall Model
RAD Model
Spiral Model
V-Model
Incremental Model
Agile Model
Iterative Model
Bigbang Model

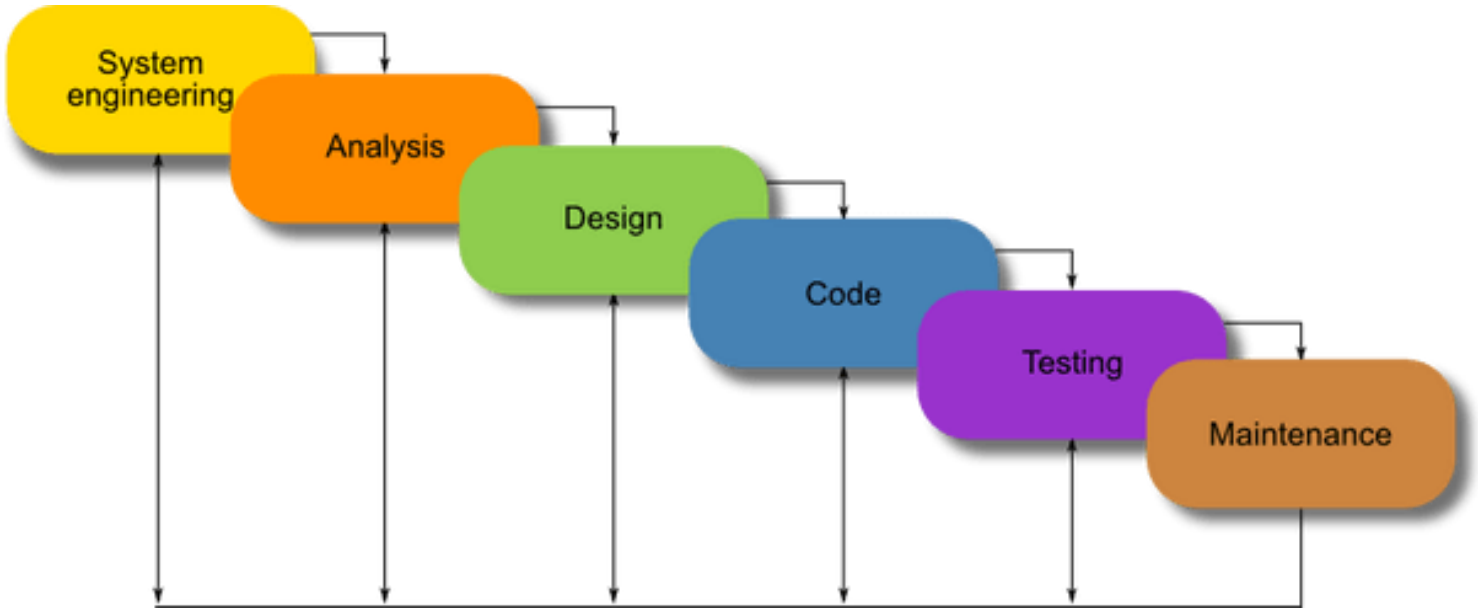
Waterfall Model



Waterfall Software Development Methodology

Waterfall methodology is a sequential design process

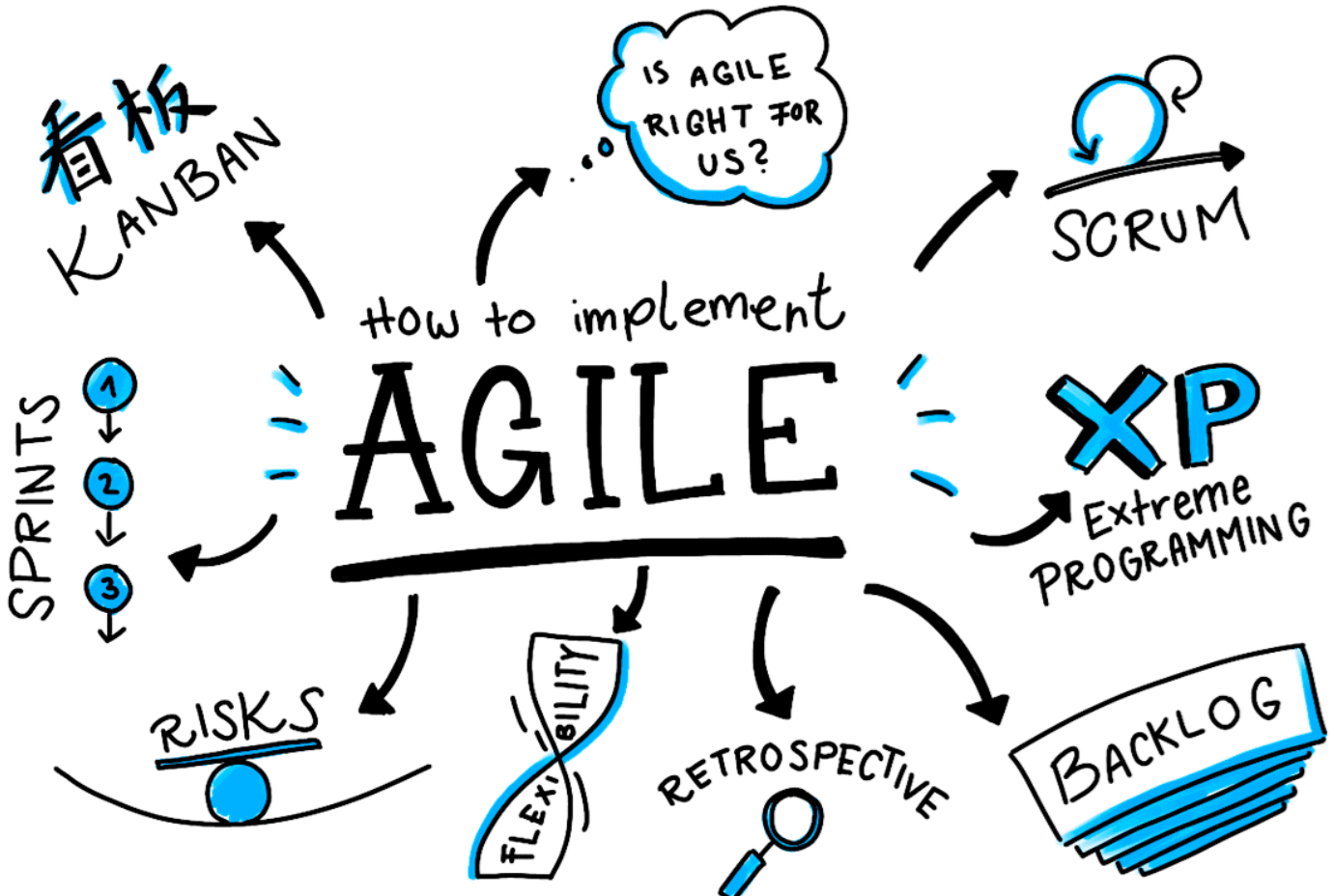
1. Conception
2. Initiation
3. Analysis
4. Design
5. Construction
6. Testing
7. Maintenance



Agile Software Development Methodology

Agile is the ability to create and respond to change

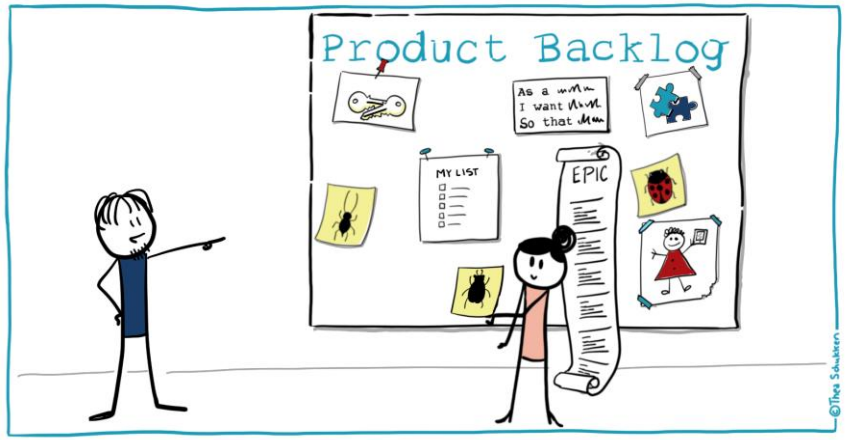
- 2001 – Agile Manifesto
- 12 Principles
- Agile is a Mindset



Key Agile Concepts

User Stories:

In consultation with the customer or product owner, the team divides up the work to be done into functional increments called “user stories.”



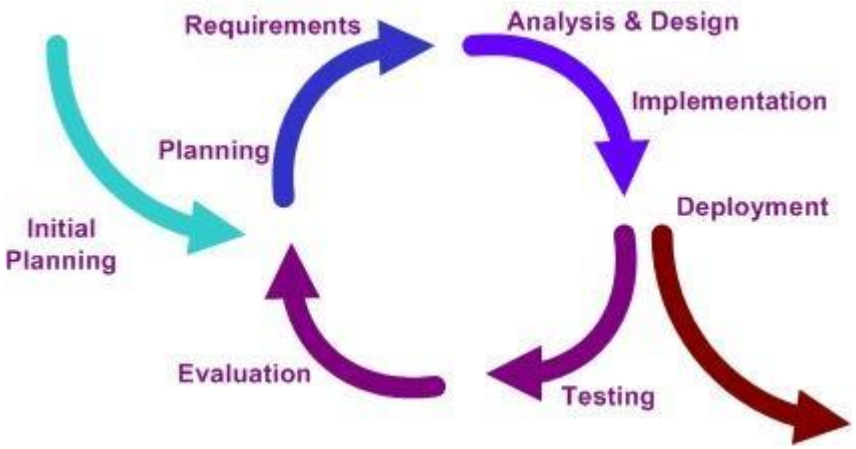
Daily meetings:

Each day at the same time, the team meets to bring everyone up to date on the information that is vital for coordination

Key Agile Concepts ..

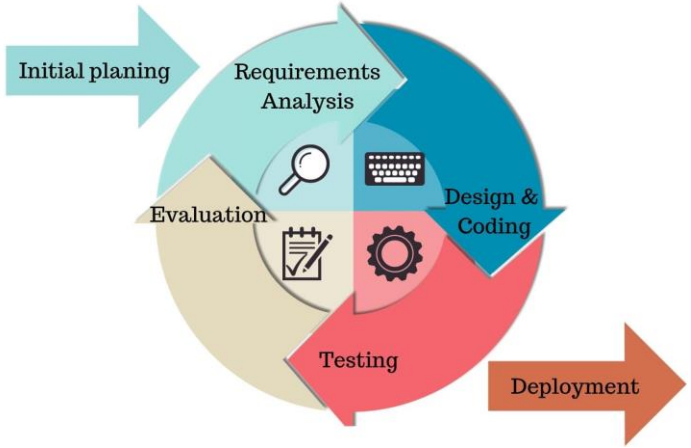
Incremental Development:

Each successive version of the product is usable, and each builds upon the previous version by adding user-visible functionality



Iterative Development:

intentionally allow for “repeating” software development activities, and for potentially “revisiting” the same work.



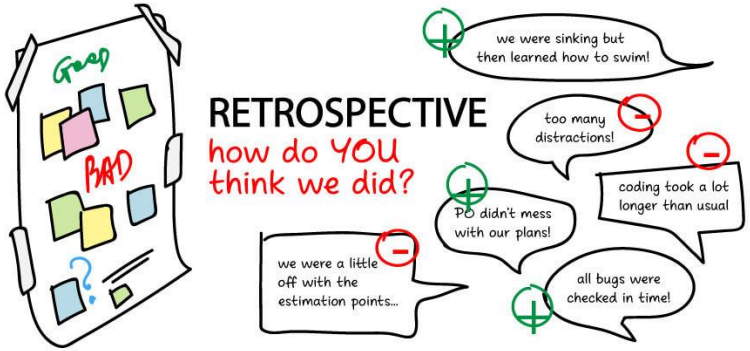
Key Agile Concepts ..

Team:

A “team” in the Agile sense is a small group of people, assigned to the same project or effort, nearly all of them on a full-time basis

Milestone Retrospective:

At the end of the project (in that case, especially when the team is likely to work together again), all the team’s permanent members (not just the developers) invests from one to three days in a detailed analysis of the project’s significant events



Key Agile Concepts ..

Personas:

"make up pretend users and design for them"

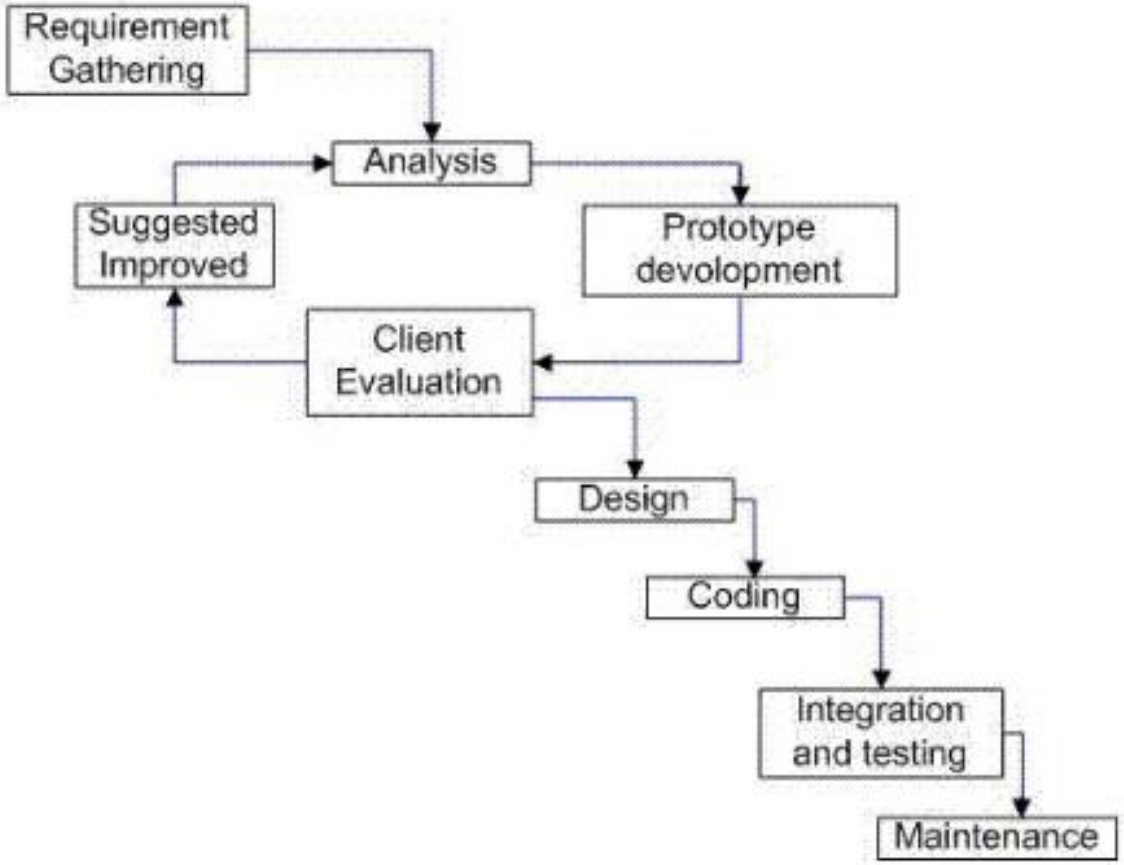
-

Alan Cooper

Software prototyping

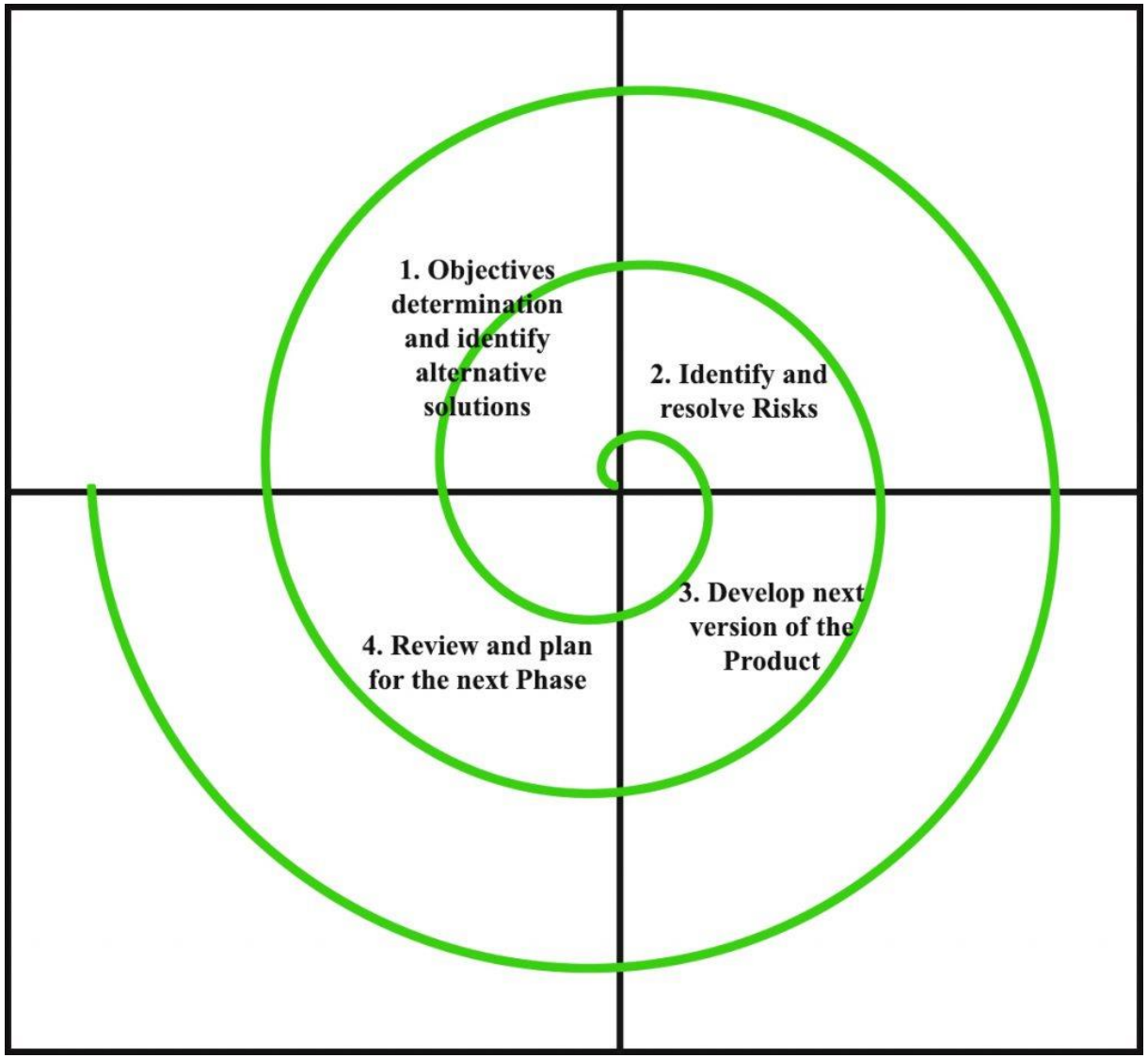
Software prototyping is the activity of creating prototypes of software applications

1. Identify basic requirements
2. Develop initial prototype
3. Review
4. Revise and enhance the prototype



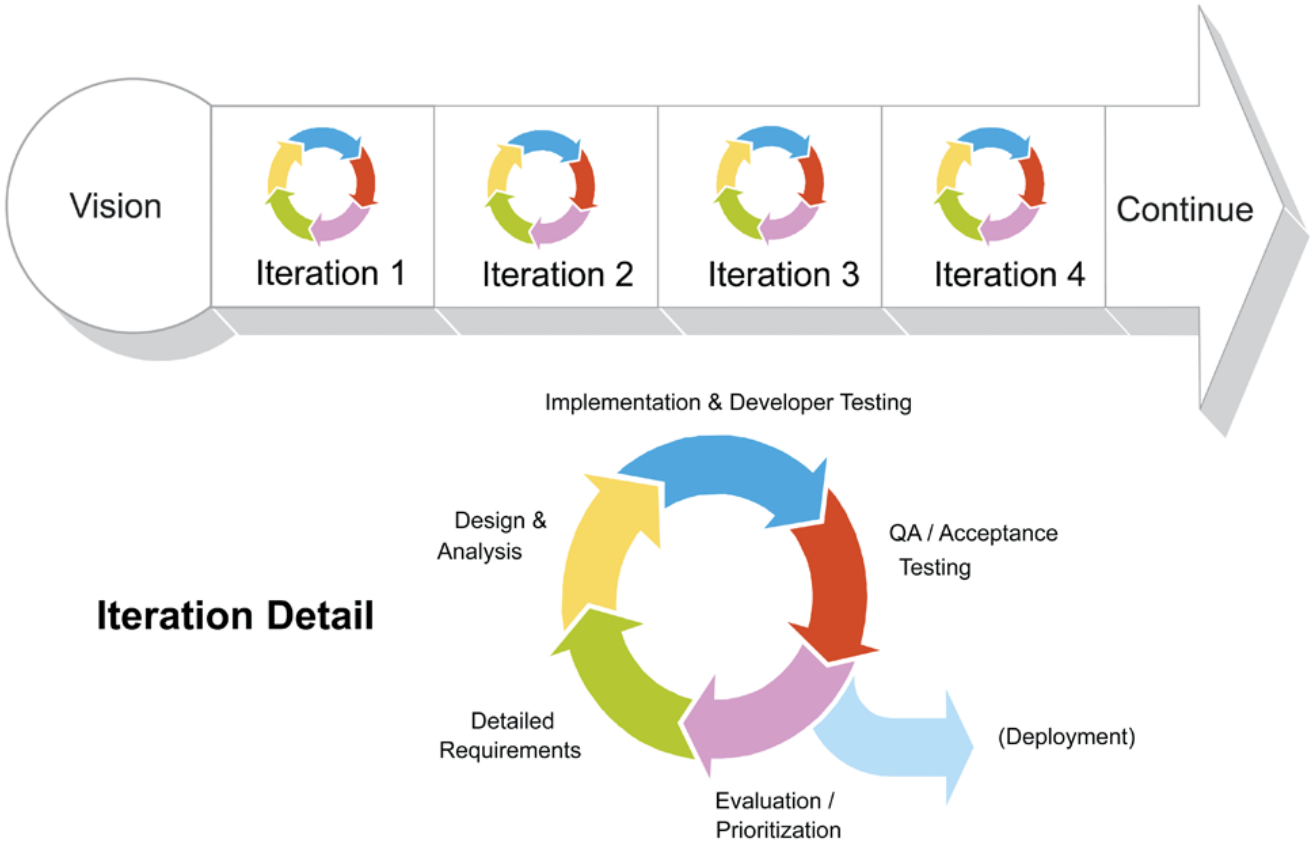
Spiral Software Development Methodology

The spiral model is a combination of Iterative development and waterfall method.



Iterative and Incremental Model

It starts with an initial planning and ends with deployment with the cyclic interactions in between



Software Requirement Specification (SRS)

SRS (Software Requirements Specification) is a document that describes the system that should be developed and implemented.

The SRS Structure:

- ❑ Introduction (purpose, scope, terms, references, overview)
- ❑ Overall description (product perspective, functions, user characteristics, assumptions, etc.)
- ❑ Specific requirements (for interface, performance, design, etc.)

SRS Contents

- Introduction (Purpose, Scope, Intended audience)
- Overall Description
  - Product Perspective (Architecture Diagram)
  - Product Functions (Use case diagram, Use case descriptions)
  - DB Requirements (E-R and DB Table structure)
  - User Classes and Characteristics (Class Diagram, Activity Diagram)
- External Interfaces
  - User Interfaces
  - Hardware Interfaces
  - Software Interfaces
  - Communication Interfaces
- Functional and Non Functional Requirements

UML Diagrams

Structural UML diagrams

- Class diagram
- Package diagram
- Object diagram
- Component diagram
- Composite structure diagram
- Deployment diagram

Behavioral UML diagrams

- Activity diagram
- Sequence diagram
- Use case diagram
- State diagram
- Communication diagram
- Interaction overview diagram
- Timing diagram

Questions????

Thank You!